

MIT Emerging Talent (ET6) – Final Reflection

Hospital Performance Analysis: Capstone Project

Arthur Dorvil | Certificate in Computer and Data Science | December 2025

Final Reflection

When I look back on this project, I realize that my greatest challenge was not technical. It wasn't Python, healthcare terminology, or correlation coefficients. The real battle was internal. Long before I struggled with dashboards or statistical analysis, I struggled with the quiet voice that kept telling me, *"You don't belong here."*

That voice followed me into MIT Emerging Talent, especially during the early milestones. It whispered when I stared at messy datasets I didn't understand. It whispered when I opened Power BI for the first time and felt lost. It whispered when deadlines approached and when my confidence felt fragile. And for a moment, I believed it because doubt had been louder than faith for most of my life.

But something changed during this project. Through the structure of the program, the support from mentors who told me, *"We didn't choose you by accident,"* and the responsibility I felt toward the work, I began confronting that voice instead of obeying it. I did not become fearless; I became brave enough to keep going even when fear was present.

Were the Original Objectives Achieved?

Yes, and in ways that exceeded my initial expectations.

I successfully analyzed hospital performance in the Bradenton–Sarasota region, cleaned and explored 119 quality measures, identified the 8-point medication communication gap, and discovered a strong negative correlation ($r = -0.92$) between patient experience and readmissions. I built a full Power BI dashboard with branded KPI cards, slicers, comparative visuals, and a narrative that connects technical findings to real-world outcomes.

But the objective evolved into something deeper.

MIT Emerging Talent (ET6) – Final Reflection

Hospital Performance Analysis: Capstone Project

Arthur Dorvil | Certificate in Computer and Data Science | December 2025

“Robert,” the patient in my story, stopped being just data. He became someone I know. He became a symbol of the people in my community who fall through the cracks, not because they are incapable, but because systems fail to explain clearly. In many ways, Robert is also a reflection of me, of the times I have been lost inside systems that didn’t speak my language.

So yes, I met the measurable objectives. But I also discovered meaning, purpose, and a reason to use data to speak for those who are often unheard.

Most Valuable Learning

The most valuable thing I learned is that data science is not just about finding patterns; it is about finding the people inside those patterns.

The moment I saw the $r = -0.92$ correlation appear in my analysis, something shifted. It wasn’t just a strong statistical relationship. It was a human truth:

When patients understand their medications, they stay out of the hospital.

This project taught me that storytelling elevates data science; it gives the numbers purpose. I learned that empathy is not a weakness in technical fields; it is a strength that shapes better questions and more meaningful insights. I learned that I am capable of teaching myself difficult concepts with patience and persistence.

Most importantly, I learned that purpose determines how far you are willing to push through difficult moments. When the project became about preventing real people from harm, every challenge became meaningful.

Project Management Journey

At the beginning, I felt completely lost. I didn’t understand the domain or the tools. But MIT’s milestone structure gave me a foothold. Breaking the project into smaller phases made the impossible feel achievable.

MIT Emerging Talent (ET6) – Final Reflection

Hospital Performance Analysis: Capstone Project

Arthur Dorvil | Certificate in Computer and Data Science | December 2025

Working alone meant I had to manage every role: data engineer, analyst, communicator, and storyteller. To stay organized, I used GitHub Projects with a Kanban board. This visual structure helped me focus on one task at a time instead of becoming overwhelmed by the full scope of the work.

Key lessons:

- Start small and iterate.
- Document everything.
- Use deadlines as discipline.
- Accept confusion as part of the process.

This project taught me that project management is not just about timelines. It is about managing your own psychology and momentum.

Approach to Team Delegation (If Redone in a Team)

If I were to redo this as a team project, I would delegate early data collection, pipeline setup, and validation to teammates who enjoy the engineering side. I would take ownership of:

- Problem formulation
- Communication strategy
- Dashboard design
- Narrative construction
- Final synthesis

These are the areas where my strengths, connecting technical analysis to other humans' meaning, create the most impact.

MIT Emerging Talent (ET6) – Final Reflection

Hospital Performance Analysis: Capstone Project

Arthur Dorvil | Certificate in Computer and Data Science | December 2025

Choosing the Public-Facing Artifact

I chose a comprehensive GitHub repository because I wanted the final product to be welcoming, accessible, and professional. The repository demonstrates the full workflow:

raw CMS data → Python cleaning → exploration → analysis → storytelling → Power BI dashboard

The Power BI dashboard was essential for healthcare stakeholders. Including hospital logos and authentic branding colors was intentional; it shows respect for each institution's identity and communicates findings in a professional, executive-ready format.

This artifact also aligns with my Healthcare IT career goals. It showcases technical proficiency, business intelligence tools, healthcare domain knowledge, communication skills, and end-to-end execution, exactly what recruiters look for (hopefully).

Most Useful Parts of the MIT ET Program

Personal Agency Skills were transformational. Learning how to manage doubt, teamwork, pressure, and emotional fatigue allowed me to continue when my confidence wanted me to quit.

The milestone structure gave me discipline.

The technical modules gave me tools.

The feedback gave me direction.

The staff gave me hope.

And the “good pressure” of MIT gave me motivation to honor the opportunity.

MIT Emerging Talent (ET6) – Final Reflection

Hospital Performance Analysis: Capstone Project

Arthur Dorvil | Certificate in Computer and Data Science | December 2025

Final Insight: Who I Became Through This Project

The most meaningful part of this journey was not the dashboard or the statistical findings. It was the version of myself I discovered.

I entered this project questioning whether I belonged in the tech world or at MIT, wondering if I had been admitted accidentally. I leave it knowing:

I belong here. I earned my place.

I am capable of far more than I believed.

I learned that I can turn raw data into something that speaks for people. Furthermore, I learned that my “Robert” story is not a weakness, but a source of power. And I learned that when I combine discipline with creativity, and data with purpose, I create work that matters.

This project didn’t just teach me data science.

It taught me identity.

It taught me purpose.

It taught me that I am not done yet. In fact, I am only getting started.

Word Count: ~960

Arthur Dorvil | MIT Emerging Talent (ET6)

Certificate in Computer and Data Science | December 2025